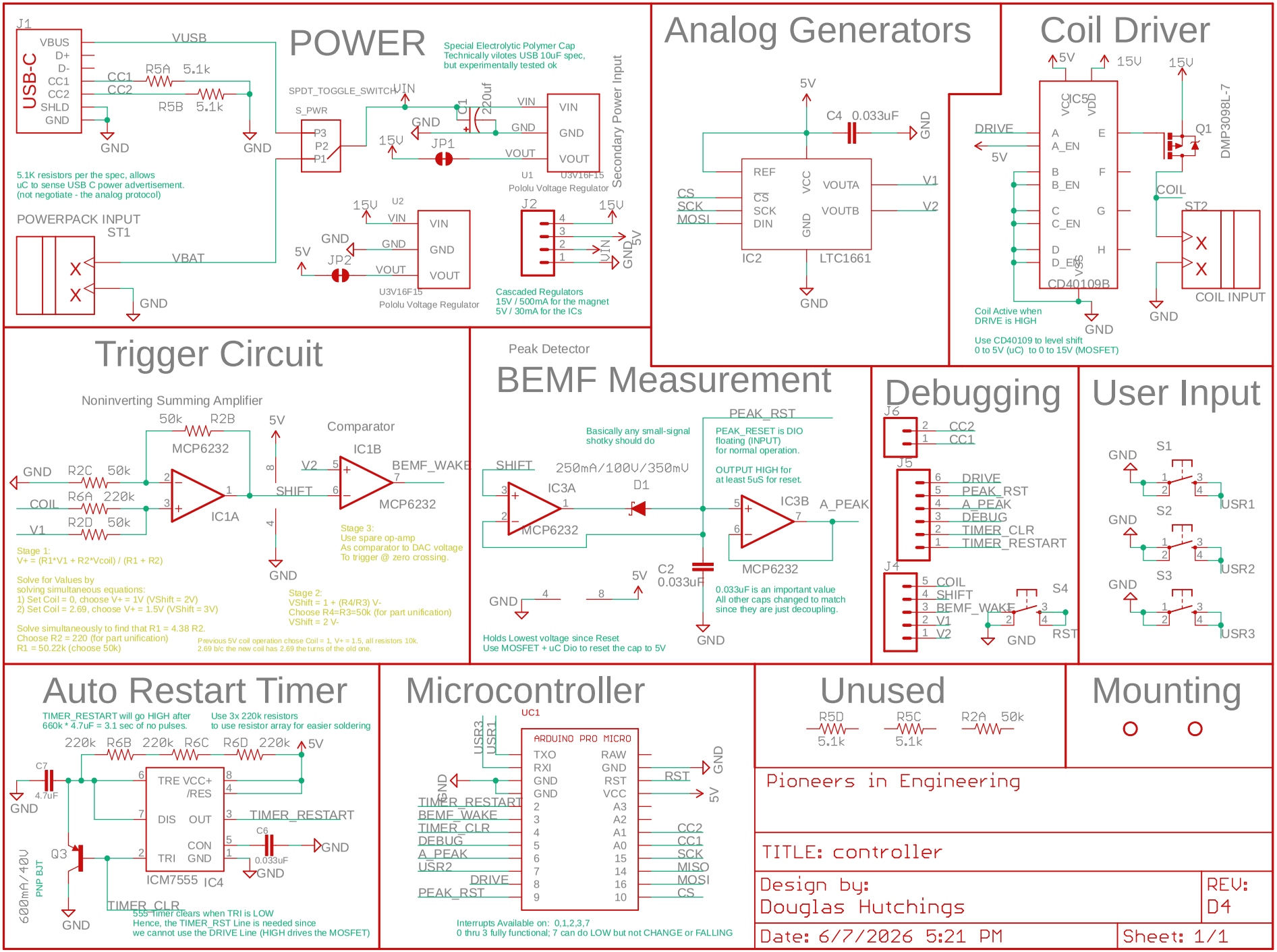
POWER
Special Electrolytic Polymer Cap
Technically vilotes USB 10uF spec,
but experimentally tested ok
5.1K resistors per the spec, allows
uC to sense USB C power advertisement.
(not negotiate - the analog protocol)
POWERPACK INPUT
Secondary Power Input
Pololu Voltage Regulator
Cascaded Regulators
15V / 500mA for the magnet
5V / 30mA for the ICs
Analog Generators
Coil Driver
Coil Active when
DRIVE is HIGH
Use CD40109 to level shift
0 to 5V (uC) to 0 to 15V (MOSFET)
Trigger Circuit
Noninverting Summing Amplifier
Comparator
Stage 1:
V+ = (R1*V1 + R2*Vcoil) / (R1 + R2)
Solve for Values by
solving simultaneous equations:
1) Set Coil = 0, choose V+ = 1V (VShift = 2V)
2) Set Coil = 2.69, choose V+ = 1.5V (VShift = 3V)
Solve simultaneously to find that R1 = 4.38 R2.
Choose R2 = 220 (for part unification)
R1 = 50.22k (choose 50k)
Previous 5V coil operation chose Coil = 1, V+ = 1.5, all resistors 10k.
2.69 b/c the new coil has 2.69 the turns of the old one.
Stage 2:
VShift = 1 + (R4/R3) V-
Choose R4=R3=50k (for part unification)
VShift = 2 V-
Stage 3:
Use spare op-amp
As comparator to DAC voltage
To trigger @ zero crossing.
Peak Detector
BEMF Measurement
Basically any small-signal
shotky should do
PEAK_RST
PEAK_RESET is DIO
floating (INPUT)
for normal operation.
OUTPUT HIGH for
at least 5uS for reset.
250mA/100V/350mV
0.033uF is an important value
All other caps changed to match
since they are just decoupling.
Holds Lowest voltage since Reset
Use MOSFET + uC Dio to reset the cap to 5V
Debugging
User Input
Auto Restart Timer
TIMER_RESTART will go HIGH after
660k * 4.7uF = 3.1 sec of no pulses.
Use 3x 220k resistors
to use resistor array for easier soldering
555 Timer clears when TRI is LOW
Hence, the TIMER_RST Line is needed since
we cannot use the DRIVE Line (HIGH drives the MOSFET)
Microcontroller
Interrupts Available on: 0,1,2,3,7
0 thru 3 fully functional; 7 can do LOW but not CHANGE or FALLING
Unused
Mounting
Pioneers in Engineering
TITLE: controller
Design by:
Douglas Hutchings
REV:
D4
Date: 6/7/2026 5:21 PM
Sheet: 1/1